Protein Synthesis

Exam code: 0610 & 0970

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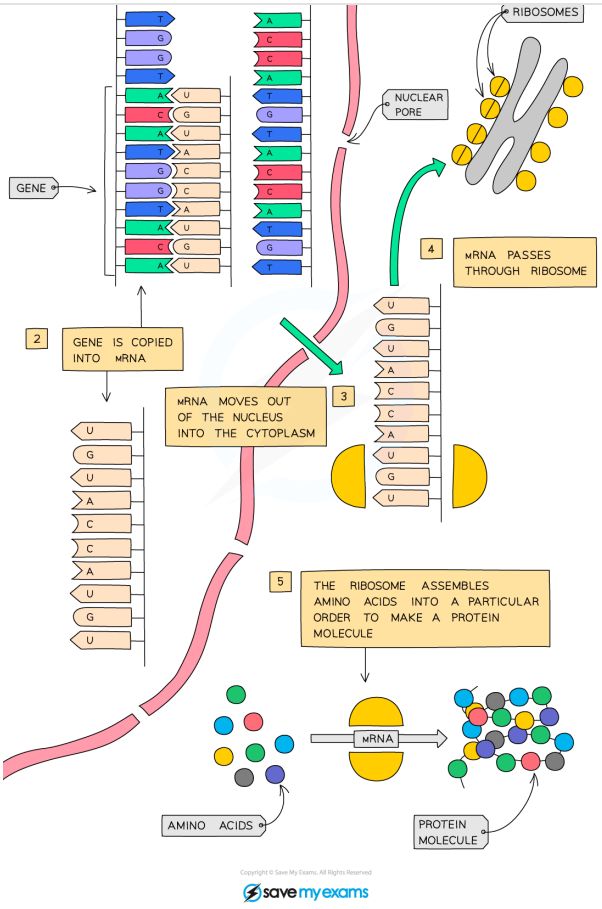
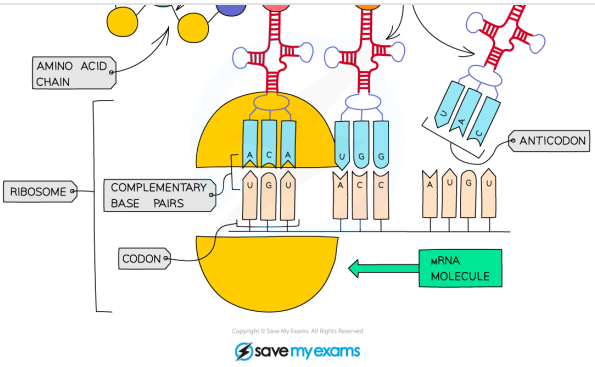


Protein synthesis

- Proteins are made by **ribosomes** with the sequence of amino acids controlled by the **sequence of bases** contained within DNA
- DNA **cannot travel out of the nucleus** to the ribosomes (it is far too big to pass through a nuclear pore) so the base code of each gene is transcribed onto an RNA molecule called **messenger RNA** (mRNA).
- mRNA then **moves out of the nucleus** and attaches to a ribosome
- The ribosome 'reads' the code on the mRNA in groups of three
- Each triplet of bases **codes for a specific amino acid**
- In this way the ribosome **translates** the sequence of bases into a **sequence of amino acids** that make up a protein
- Each tRNA molecule binds to, and carries, a single, specific amino acid from the cytoplasm to the ribosome.for the assemby of the amino acid chain
- Once the amino acid chain has been assembled, it is released from the ribosome so it can fold and form the final structure of the protein



The triplet code of DNA (carried by mRNA) is read by the ribosome and amino acids are attached together in a specific sequence to form the protein

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